Dynamic Recurrent Self-Refinement Network for Hyperspectral Remote Sensing Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution (HSISR) is crucial for enhancing the spatial detail of hyperspectral images (HSIs). However, existing single HSISR methods face challenges with the effective use of spectral context and generalization across varying spectral bands. We propose a dynamic recurrent self-refinement network (DRSN) to address these limitations. DRSN innovatively models the HSISR process as a nonlinear dynamic system evolving along the spectral dimension, where the spectral groups are treated as state variables. This formulation inherently allows DRSN to process HSIs with arbitrary band counts, thereby broadening the application scope of HSISR. The key contributions in DRSN can be summarized into two novel modules: adaptive state predicting and updating (ASPU) and bidirectional cross-state alignment (BCSA). The ASPU employs an uncertainty-guided adaptive activation mechanism to dynamically refine the current state by selectively integrating interstate complementary information while suppressing irrelevant context. The BCSA utilizes an efficient back-and-forth caching strategy and masked intra- and interstate attention (MIISA) to mitigate information imbalance and achieve effective bidirectional contextual alignment without substantial computational overhead. Experiments on benchmark HSI datasets demonstrate that DRSN achieves state-of-the-art (SOTA) performance, exhibits wide applicability, and maintains a lightweight and computationally efficient structure. The source code is available at https://github.com/hgpftd/DRSN**

***Index Terms*—Dynamic recurrent network, hyperspectral image super-resolution (HSISR), remote sensing, spectral correlation.**

I. Introduction

HYPERSPECTRAL remote sensing image could describe various kinds of characteristics on the Earth's surface from different bands of the electromagnetic spectrum [1], which provides sufficient spectral and spatial structural information to support further practical applications and analyses. In recent years, hyperspectral images (HSIs) have been widely applied in plant detection [2], urban analysis [3], and disaster detection [4]. However, during the imaging process, dense spectral bands restrict incident energy to reach the sensor, resulting in a relatively low spatial resolution of HSIs and limiting subsequent applications. Thus, there is often a tradeoff between spatial resolution and spectral resolution.

HSI super-resolution (HSISR) [5] aims to reconstruct a high-resolution (HR) HSI from the corresponding low-resolution (LR) counterpart. It can improve the spatial resolution of HSIs without changing the imaging devices, which has attracted significant attention from researchers in the field of remote sensing. Existing HSISR methods are broadly categorized into single-image SR [6], [7], [8] and multi-image SR [9], [10], [11], distinguished by their reliance on auxiliary information. Among these methods, multi-image SR approaches usually employ a fusion-based strategy to reconstruct the target HR HSI by combining the LR HSI with auxiliary HR images (e.g., RGB images). Despite their promising performance, the dependence on strictly co-registered auxiliary images with the LR HSI severely curtails their practical applicability.

In contrast, single-image SR reconstructs HR HSIs using only the LR HSIs without additional auxiliary information, which renders these methods more promising for practical scenarios. Traditionally, most single-image SR methods rely on spatial or spectral statistical distributions to reconstruct HR HSIs, such as sparse representation [12] and low-rank decomposition [13]. However, due to the inherent high dimensionality of HSIs, these methods often fail to adequately capture the complex intrinsic data characteristics, consequently yielding unsatisfactory reconstruction results. With the widespread application of deep learning in the field

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of computer vision, numerous deep learning-based HSISR methods have been proposed, demonstrating promising performance. Approaches based on the convolutional neural network (CNN) [14], [15] usually employ 2-D/3-D convolutions to extract image features, which achieve superior performance compared to traditional methods. However, convolutions primarily capture spatial features while overlooking crucial spectral information, which can lead to spectral distortion. Recently, several Transformer-based methods have been proposed [16], [17], as Transformers can effectively learn global information through the attention mechanism. These methods achieve better reconstruction performance compared to previous approaches. However, most of these methods simultaneously input all bands into the Transformer backbone to conduct self-attention, which makes them struggle with the high data dimensionality of HSIs. This high dimensionality increases the difficulty of feature extraction and imposes significant computational costs. Moreover, high-dimensional HSI data contain complex spectral correlations, which are often neglected by these methods, thereby leading to an underutilization of the complementary information between spectral bands.

To address these challenges, several grouping-based methods have been proposed. These methods divide an HSI into several groups along the spectral dimension for individual processing [6], [18], [19], and then fuse them to form the output. In this formulation, each group is treated as an independent unit to extract spatial–spectral features. The intermediate results of each group are subsequently merged to further extract deep features. With these procedures, they successfully reduce the spectral dimensionality to alleviate the negative impact of HSI's high dimensionality on reconstruction. In addition, they can effectively exploit the spectral similarity between adjacent bands inside each group.

To further exploit the complementary information between groups, some researchers adopt recurrent-based methods to process spectral groups, which can be categorized into unidirectional and bidirectional recurrent structures [8], [20], [21], [22]. Although these methods show progress in exploiting interband spectral correlations, several challenges remain unsolved. First, hyperspectral datasets typically vary in the number of spectral bands. Most existing models can only handle HSIs with a fixed number of spectral bands as those in the training set, thus failing to process HSIs with an arbitrary number of spectral bands. This disadvantage significantly limits the applicability of these methods. Second, most current methods overlook the complementarity and difference between spectral groups. Although the HSI is divided into several spectral groups to mitigate the high-dimensional challenge, all groups are usually treated equally during the group fusion process. In fact, the reflectance of the same ground object can vary greatly across different bands, resulting in a contextual bias where later groups benefit from more accumulated information than earlier ones. As a result, fusing subgroups in this manner may introduce irrelevant context captured from uninformative bands and fail to adaptively focus on beneficial complementary information between groups. Finally, existing recurrent-based methods face a dilemma. The unidirectional recurrent structure propagates only the results of previous groups to the current group for reconstruction, which leads to an imbalance of information between different spectral groups. Although the bidirectional recurrent structure can combine information flows from forward and backward directions, it introduces significant computational costs.

In this article, we propose a novel HSISR approach to solve these challenges. For the first issue, we propose a dynamic recurrent self-refinement network (DRSN) based on system dynamics. To fully exploit the spectral continuity inherent in HSIs, we formulate the HSIs as a nonlinear dynamic system evolving along the spectral dimension, with each spectral group treated as a state variable of the system. This dynamic system iteratively models HSI data from each spectral group and updates the corresponding state variable, which allows our approach to pause or proceed at any given spectral band, thereby breaking the constraints on the HSI spectral band numbers. For the second issue, we carefully design an adaptive state predicting and updating (ASPU) module to leverage the complementary context between adjacent states to dynamically refine the current state. During the iteration of the dynamic system, this module continuously collects the state of each band and predicts the current state based on previous states. Based on uncertainty estimation, the adaptive state activation mechanism (ASAM) in ASPU adaptively activates valuable spatial–spectral information within the adjacent state, which is encoded from the corresponding spectral group. It integrates this cross-state context while effectively excluding irrelevant interference. For the third issue, we propose a bidirectional cross-state alignment (BCSA) module to further align each state through cross-state information. It utilizes the back-and-forth caching strategy to enable bidirectional information propagation under a unidirectional recurrent structure. The masked intra- and interstate attention (MIISA) is employed to achieve efficient cross-state information interaction through an adaptive mask generated from spectral difference. Due to these two designs, BCSA mitigates the information inconsistency presented in a unidirectional recurrent structure without the need for large memory.

In summary, the main contributions of this work can be summarized as follows.

1) We propose a novel HSISR approach DRSN, reformulating the HSI as a state variable sequence from a nonlinear dynamic system perspective. This allows our approach to handle HSIs with varying numbers of bands, even those differing from training data, achieving better applicability.
2) We develop an ASPU to perceive differences between states and dynamically adjust and refine the spectral states accordingly. It enables the network to adaptively focus on beneficial complementary information through the ASAM.
3) The BCSA utilizes a back-and-forth caching strategy for bidirectional context acquisition without requiring large memory, which is employed along with MIISA for adaptive cross-state interaction without introducing irrelevant information. BCSA enables efficient

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bidirectional information propagation within a unidirectional recurrent structure.

4) Extensive experiments on multiple hyperspectral remote sensing image datasets demonstrate the superiority and generality of DRSN, e.g., DRSN outperforms the second-best method by 0.19 dB in peak signal-to-noise ratio (PSNR) on Houston (×4) even with only half the computational cost.

II. Related Work

A. Single Natural Image SR

Deep learning-based methods have achieved excellent performance in single natural image SR tasks. As pioneering work in this field, Dong et al. [23] developed a three-layer CNN architecture SRCNN to learn the mapping from LR images to HR images, demonstrating superior performance compared to traditional algorithms. Building on the success of residual learning for stabilizing and deepening networks, many CNN-based deep networks have been proposed for better performance, such as FSRCNN [24], VDSR [25], EDSR [26], and SRDenseNet [27]. Subsequently, some methods improved network performance through the attention mechanism. For instance, RCAN [28] introduced a channel attention (CA) mechanism, while CSNLN [29] designed effective spatial attention. Recently, Transformers have achieved significant success in image SR tasks, such as SwinIR [30] and Swin2SR [31], effectively extracting global features of images. In order to reduce computational overhead, Lu et al. [32] proposed an efficient super-resolution Transformer (ESRT) through a specially designed lightweight multi-head attention mechanism. HAT [33] combined CA with overlapping and nonoverlapping self-attention mechanisms to activate more pixels for better reconstruction. Considering the complementarity between Transformers and CNNs, some dual-branch hybrid networks [34], [35] were designed to leverage global and local features. Moreover, researchers also explored the generalization of networks in real-world scenarios through strategies such as dictionary learning [36], multitask learning [37], and test-time adaptation [38].

Despite their remarkable success on natural images, most of them are not directly applicable to HSISR. This is primarily because they are designed to recover spatial details and often overlook the crucial spectral information inherent in HSIs. Such neglect can lead to significant spectral distortion in the reconstructed HSIs. Furthermore, these methods lack the ability to utilize complementary information between spectral bands, thus failing to fully exploit the rich spectral information inherent in HSIs.

B. Single HSISR

Single HSISR has recently garnered significant attention due to its wide practicality. Yuan et al. [39] utilized the transfer learning technique and employed nonnegative matrix factorization to strengthen the constraints on the results. Similarly, Xie et al. [40] combined the CNN with nonnegative matrix factorization to obtain HR images. However, these early methods often require complex constraints on the spatial–spectral structure of intermediate reconstruction results, rather than performing end-to-end HSISR.

Recently, end-to-end methods have gradually become the mainstream to simplify the reconstruction process and better utilize spatial–spectral features. Li et al. [14] proposed SDCNN to learn the end-to-end spectral difference mapping between the LR and HR images. Liu et al. [15] incorporated the structural priors of HSIs to achieve excellent performance with fewer parameters. However, these 2-D convolution-based methods primarily capture spatial features while overlooking spectral information, which may lead to serious spectral distortion. To better exploit spatial–spectral information, several methods based on 3-D convolution have been proposed. Li et al. [41] extracted spatial and spectral features separately by investigating hybrid structures that combine 2-D and 3-D convolutions. SRDNet [42] employed hybrid convolutions to capture interspectral self-similarity in the spatial domain. Jia et al. [43] introduced spectral convolutions into the enhanced convolution block to better learn spectral information. To overcome the limited receptive field of CNNs, researchers have introduced Transformers into the HSISR domain to model global spatial–spectral dependencies [44], [45], [46], yet the heavy computational cost of self-attention mechanisms often poses a significant burden. To address this efficiency bottleneck, state-space models (SSMs) with contextual spatial–spectral scanning [47], [48] have been employed for linear-complexity dependence modeling. However, the above methods extract features by simultaneously processing all spectral bands, which lacks exploration of spectral correlation in reconstruction.

To better exploit spectral correlations and manage high dimensionality, a common strategy involves spectrally partitioning HSIs into several groups, which are often initially processed independently before potential fusion. For instance, Liu et al. [19] utilized group convolutions to extract features from each group and introduced a spectral attention mechanism to focus on the exploration of spatial–spectral features. Li et al. [49] explored a two-stage network that reconstructs HSI group-by-group and then learns to enforce spatial–spectral consistency. Wang et al. [6] proposed a group embedding learning method and enriched spectral complementary information through the integration of neighboring groups. MSDFormer [7] used CNN to process the local information of each group, and then merged the groups and employed a Transformer to extract global information. SCPSN [50] introduced a spectral clustering-based grouping strategy to aggregate similar bands and exploited symmetric groups for complementary information. However, the isolated processing of spectral groups in these methods often limits intergroup interaction. This can lead to an underutilization of spectral continuity across group boundaries and may potentially disrupt the global spatial–spectral consistency of the HSI.

More recently, several methods based on recurrent feedback structures have been proposed to extract continuous spectral information by using features from previous bands as feedback for the current band. For instance, Fu et al. [21] proposed a bidirectional 3-D quasi-recurrent network to learn complementary information more effectively.

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DQSR [22] proposed a bidirectional recurrent network that develops different feature extraction units for nonadjacent and adjacent bands. RFSR [20] developed a unidirectional recurrent feedback network to explore interband correlations and maintained spatial–spectral consistency through a regularization network. HCMN [51] introduced a hierarchical context measurement network that calculates feature and pixel similarities to align and fuse contextual features from adjacent bands. Wang et al. [8] introduced a two-stage recurrent network to model the relationship between adjacent bands, enhancing the complementarity of contextual information. However, existing recurrent-based methods face a dilemma. The unidirectional recurrent structure leads to an imbalance of information between different spectral bands. Although the bidirectional recurrent structure can utilize bidirectional information, it introduces significant computational costs. Furthermore, some methods overlook the correlations and differences between the feedback information and the current band. All components are treated equally when integrating feedback information with the current band, which may introduce irrelevant context and fail to adaptively focus on beneficial complementary information.

Large HSI models have emerged as a powerful paradigm for HSI processing. HyperSIGMA [52] established a billion-parameter transformer using sparse sampling attention to learn universal spectral representations. HyperFree [53] introduced channel-adaptive embeddings to enable tuning-free processing of HSIs with varying bands. Meanwhile, diffusion-based generative models have been applied to HSISR. DMGASR [54] and SSAT [55] explored efficient representations by performing diffusion in low-dimensional latent space or leveraging pretrained diffusion priors, respectively. ISPDiff [56] and HDiff-HIR [57] focused on enhancing reconstruction fidelity by incorporating physically interpretable models or hierarchical conditions into the iterative denoising process. However, these methods generally incur heavy computational burdens due to massive parameter counts or iterative diffusion inference.

In this work, we propose a DRSN to perceive differences between spectral bands and dynamically adjust the current bands accordingly, which enables the network to adaptively focus on valuable complementary information between spectral bands and exclude the interference of irrelevant information. In addition, we introduce a BCSA module to utilize bidirectional spectral information, allowing our network to achieve the efficiency of unidirectional networks while enabling bidirectional information propagation.

III. Proposed Methodology

In this section, we first establish a theoretical formulation for HSISR from a dynamic system perspective. Subsequently, we introduce the overall architecture of the proposed DRSN and elaborate on its two core components: the ASPU module and the BCSA module.

A. Problem Formulation

The high spectral continuity in HSIs implies that the features of one spectral group can be seen as a natural evolution from the preceding one. This inherent evolutionary characteristic aligns well with the principles of nonlinear dynamic systems, motivating us to model the HSISR task as a recursive state estimation process that evolves along the spectral dimension. Specifically, let the target HR HSI be a sequence of spectral groups $Y = \{y_g\}_{g=1}^{G}$. The *state equation* is defined as

$$y_g = \mathcal{F}\left(y_{g-1}\right) + q_g \tag{1}$$

where y_g is the state of gth spectral group, $\mathcal{F}(\cdot)$ is the nonlinear state transition function, and q_g is the process noise that accounts for unpredictable spectral variations. The relationship between the observed LR data x_g and the true HR state y_g is described by the *observation equation*

$$x_g = \mathcal{D}\left(y_g\right) + r_g \tag{2}$$

where $\mathcal{D}(\cdot)$ is the degradation function (e.g., bicubic downsampling) and r_g is the measurement noise. Based on the formulation above, the *evolutionary principles* of this dynamic system can be defined by two recursive steps: *state prediction* and *state update*. In this work, all computations are performed in a high-dimensional state space, where x_g is first mapped to an observation state S_g via an encoder $\mathcal{E}(\cdot)$

$$S_g = \mathcal{E}\left(x_g\right). \tag{3}$$

Based on the correlation between adjacent bands, the state prediction step estimates the prior state of the current step, S_g^-, from the posterior state of the previous step, S_{g-1}^+. Unlike standard dynamic systems that predict solely from S_{g-1}^+, we additionally incorporate the current and past observations (S_g, S_{g-1}) as control inputs to perceive observational trends for more reliable prediction. This process is formulated as

$$S_g^- = \mathcal{F}_{trans}\left(S_{g-1}^+, S_g, S_{g-1}\right) \tag{4}$$

where $\mathcal{F}_{trans}(\cdot)$ represents the state transition function, which is conditioned on the current and past observations. Next, the state update step aligns the new observation with the prior prediction to obtain the final posterior state

$$S_g^+ = \mathcal{K}_g \cdot S_g + \left(1 - \mathcal{K}_g\right) S_g^- \tag{5}$$

where $\mathcal{K}_g$ is the adaptive gain that balances the prior prediction and current observation. In standard dynamic systems, the gain $\mathcal{K}_g$ is primarily driven by the prediction uncertainty $U_{g,g-1}$, which is characterized by the predicted error covariance matrix. Due to the intractability of covariance estimation for high-dimensional HSIs, we adopt a data-driven approach to implicitly learn the mappings from the states (S_g, S_{g-1}) to $U_{g,g-1}$, and subsequently from $U_{g,g-1}$ to $\mathcal{K}_g$, which is formulated as

$$\mathcal{K}_g = \mathcal{F}_{act}\left(\mathcal{F}_{un}\left(S_g, S_{g-1}\right)\right) \tag{6}$$

where $\mathcal{F}_{un}(\cdot)$ is the parameterized uncertainty estimator, and $\mathcal{F}_{act}(\cdot)$ is the mapping network that generates the final gain.

The aforementioned formulation models the HSISR task as a dynamic system evolving along the spectral dimension. Inspired by this, we parameterize a network to implement this recursive self-refinement process.

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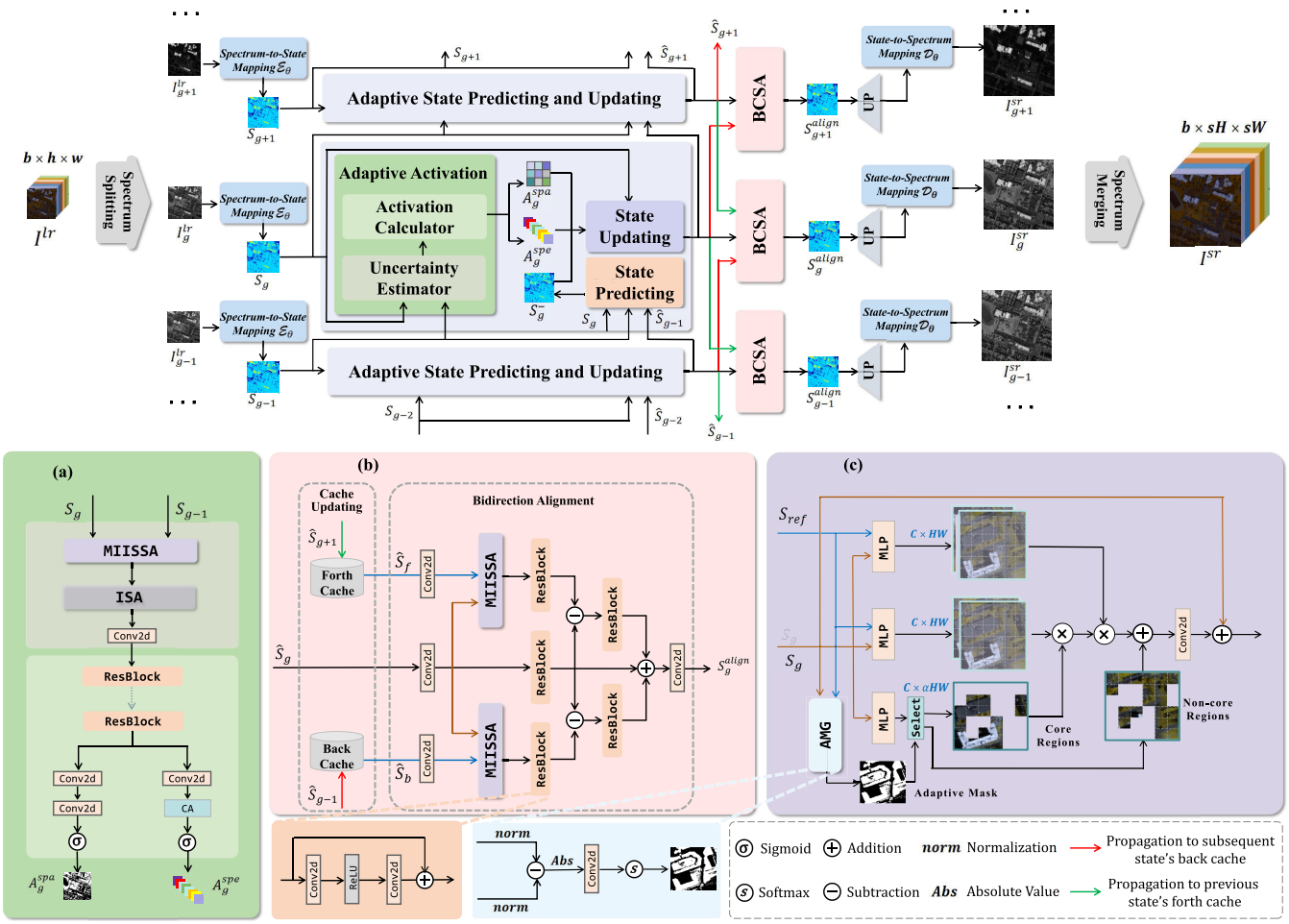


Fig. 1. Overview of the DRSN and the core components. DRSN consists of two core components: the ASPU module and the BCSA module. (a) Adaptive activation. (b) BCSA. (c) MIISA.

B. Overall Architecture

Let $I^{lr} \in \mathbb{R}^{H\times W\times B}$ denote the LR HSI, where H, W, and B represent the height, width, and number of spectral bands, respectively. For the HSISR task, the objective is to reconstruct the corresponding HR HSI $I^{sr} \in \mathbb{R}^{sH\times sW\times B}$ from I^{lr}, where s represents the upscale factor.

Initially, the input HSI I^{lr} is spectrally partitioned into G nonoverlapping groups $\{I_g^{lr}\}_{g=1}^{G}$, where $I_g^{lr} \in \mathbb{R}^{H\times W\times \frac{B}{G}}$ denotes the gth group. We leverage the proposed DRSN to reconstruct these consecutive groups sequentially, which allows us to fully exploit the continuity and variability across the spectral bands. Specifically, as shown in Fig. 1, the framework of DRSN consists of two core components: the ASPU module and the BCSA module.

In this work, we reformulate the HSIs from a nonlinear dynamic system perspective, with each spectral group treated as a state variable of the system. A key advantage of this dynamic system formulation is its inherent flexibility regarding the number of spectral bands. Since the network processes the state sequence iteratively, its core architecture is agnostic to the sequence's total length. This allows DRSN to naturally handle HSIs with arbitrary band counts, which broadens its applicability across diverse hyperspectral sensors. For the gth spectral group I_g^{lr}, we first project it into the state space

$$S_g = \mathcal{E}_\theta\left(I_g^{lr}\right) \tag{7}$$

where $\mathcal{E}_\theta(\cdot)$ is a 2-D convolution with a kernel size of 3×3, $S_g \in \mathbb{R}^{H\times W\times C}$ is the corresponding state representation of I_g^{lr}, and C is the dimensionality of the state space. Next, the state representation S_g, encoded from the current spectral group, is fed into the proposed ASPU module for further refinement.

Since our network performs reconstruction iteratively along the spectral dimension, ASPU can naturally utilize information from the preceding group to reconstruct S_g into a refined state $\hat{S}_g$ with richer details. Specifically, S_g is fed into ASPU along with the previous state S_{g-1} and the refined previous state $\hat{S}_{g-1}$ to obtain $\hat{S}_g$. As a high-quality reference, $\hat{S}_{g-1}$ may contain textures and details missing in S_g. The ASPU utilizes the spatial–spectral uncertainty estimated from S_g and S_{g-1} to guide the adaptive activation of valuable context from $\hat{S}_{g-1}$ for refining the current state S_g. This is formulated as

$$\hat{S}_g = \mathcal{ASPU}\left(S_g, S_{g-1}, \hat{S}_{g-1}\right) \tag{8}$$

where $\hat{S}_g \in \mathbb{R}^{H\times W\times C}$ is the refined current state.

Although the ASPU effectively refines each state, its operation within a unidirectional framework leads to an imbalanced information flow. To address this issue, we propose a BCSA

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module. Specifically, BCSA employs a back-and-forth caching strategy to temporarily store the refined states from preceding and succeeding spectral groups, which are generated by the ASPU. BCSA, then, aligns the refined current state $\hat{S}_g$ with these two cached states to incorporate bidirectional context. The aligned result of the gth state $S_g^{align} \in \mathbb{R}^{H\times W\times C}$ can be obtained by

$$S_g^{align} = \mathcal{BCSA}\left(\hat{S}_b, \hat{S}_g, \hat{S}_f\right) \tag{9}$$

where $\hat{S}_b$ and $\hat{S}_f$ are the cached refined states from the backward and forward directions, respectively. Subsequently, the aligned state is first upsampled and then decoded to obtain the SR result of the corresponding spectral group

$$I_g^{sr} = \mathcal{D}_\theta\left(Up\left(S_g^{align}\right)\right) \tag{10}$$

where $I_g^{sr} \in \mathbb{R}^{sH\times sW\times \frac{B}{G}}$ is the SR result of the gth group, $\mathcal{D}_\theta(\cdot)$ denotes a 2-D convolution with a kernel size of 3×3, and $Up(\cdot)$ is an upsampling module that increases the spatial dimensions by a factor of s.

Finally, the SR results of all groups will be concatenated along the spectral dimension to form the final SR HSI. To reduce the difficulty of training, the global residual is added to the result. The final reconstruction I^{sr} can be expressed as follows:

$$I^{sr} = Concat[I_1^{sr}, \ldots, I_g^{sr}, \ldots, I_G^{sr}] + I^{lr}\uparrow \tag{11}$$

where $I^{lr}\uparrow \in \mathbb{R}^{sH\times sW\times B}$ represents the original LR image I^{lr} upsampled by the factor of s using bicubic interpolation.

C. Adaptive State Predicting and Updating

In Section III-A, we reformulate HSIs as a nonlinear dynamic system evolving along the spectral dimension. In this dynamic system, a key challenge is how to effectively leverage the previous state to predict and enhance the current state. In the HSISR task, not all information in the adjacent bands is beneficial for reconstructing the current bands due to variations in reflectance. Therefore, directly fusing the previous state with the current state without suppression mechanisms may result in the propagation of redundant or irrelevant information.

To address this issue, the ASPU module, parameterized based on the formulation in Section III-A, is proposed to adaptively activate the previous state information to refine the current state without introducing interference context. The workflow of the proposed ASPU is shown in Fig. 1. Specifically, ASPU consists of three components: adaptive state activation, state predicting, and state updating (SU). In the following, we elaborate on these components and the proposed masked intra- and interstate attention (MIISA).

1) Adaptive State Activation Mechanism: Considering the points discussed above, it is desirable for the system to perceive both the differences and correlations between adjacent states. Moreover, the system should be able to adaptively activate valuable information from the previous state based on difference perception to enhance the current state. Motivated by this, we propose an ASAM, which consists of two components: an uncertainty estimator and an activation map calculator.

ASAM is guided by the principle of predicting spatial–spectral uncertainty, which physically represents the degree of prediction degradation caused by spectral variations between adjacent states. In standard dynamic systems, this uncertainty is explicitly characterized by a predicted error covariance matrix. However, direct covariance estimation is intractable for high-dimensional HSIs. Therefore, we employ a parameterized uncertainty estimator to implicitly learn this uncertainty from the previous and current states. Specifically, we propose the MIISA to fuse information from adjacent states. MIISA can fully perceive the spatial–spectral differences between adjacent states via an adaptive mask mechanism when integrating cross-state information.

In order to further explore spatial–spectral information within the fused state, we employ an intrastate attention (ISA) module $f_{isa}(\cdot)$. Here, ISA is implemented as a standard self-attention mechanism [58] to capture the internal dependencies of the fused state. Finally, the spatial–spectral uncertainty $U_{g,g-1} \in \mathbb{R}^{H\times W\times C}$ is obtained through a 1×1 2-D convolution. The workflow of the uncertainty estimator could be expressed as follows:

$$U_{g,g-1} = f_{conv}\left(f_{isa}\left(f_{miisa}\left(S_g, S_{g-1}\right)\right)\right) \tag{12}$$

where S_g and S_{g-1} are the current and previous states, respectively, and $f_{miisa}(\cdot)$ is the proposed MIISA.

In order to adaptively activate information to refine the current state, we propose an activation calculator to generate an adaptive activation mask based on the obtained spatial–spectral uncertainty $U_{g,g-1}$. The adaptive activation mask activates regions in adjacent states with lower uncertainty while suppressing those with higher uncertainty. The workflow of the proposed activation calculator is shown in Fig. 1(a).

In the activation calculator, cascaded U-shaped residual blocks are first applied to learn the multiscale data distribution. Since both spatial and spectral information are crucial for HSIs, the theoretical gain term K_g is instantiated in practice by decoupling it into the spatial and spectral adaptive activation masks, which are produced by a two-branch structure. Specifically, the first branch consists of a 3×3 and a 1×1 2-D convolution followed by a sigmoid activation function to generate spatial activation $A_g^{spa} \in \mathbb{R}^{1\times H\times W}$. The second branch includes a 3×3 2-D convolution and a CA block to produce spectral activation $A_g^{spe} \in \mathbb{R}^{C\times 1\times 1}$. Here, H, W, and C represent the height, width, and number of channels, respectively. The above process can be represented as follows:

$$A_g^{spa} = \sigma\left(f_{conv_2}\left(f_{conv_1}\left(f_{res}\left(U_{g,g-1}\right)\right)\right)\right) \tag{13}$$

$$A_g^{spe} = \sigma\left(f_{CA}\left(f_{conv}\left(f_{res}\left(U_{g,g-1}\right)\right)\right)\right) \tag{14}$$

where $\sigma(\cdot)$ is the sigmoid activation function, $f_{res}(\cdot)$ is composed of the cascaded residual blocks, and $f_{CA}(\cdot)$ represents the CA. The CA first obtains channelwise statistics by global average pooling (GAP), followed by two 1×1 convolutional layers and an intermediate rectified linear unit (ReLU).

Due to the above ASAM, the proposed method could fully perceive the differences and correlations between adjacent states. Consequently, it enables the selective utilization of information from neighboring states via the adaptive spatial and spectral activation masks.

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2) Masked Intra- and Interstate Attention: Due to the spatial heterogeneity of ground object reflectance, the valuable complementary information between adjacent states, which are encoded from spectral groups, is typically concentrated in specific spatial regions. This physical property motivates a more intelligent fusion strategy than standard attention. Therefore, we propose an MIISA mechanism, as illustrated in Fig. 1(c). Instead of performing attention globally, MIISA first utilizes an adaptive mask to pinpoint these informative regions. It, then, focuses the attention computation exclusively on these masked areas.

The proposed MIISA is designed to refine the current state S_g by adaptively incorporating context from a single reference state S_{ref}, achieving an information flow from S_{ref} to S_g. Within MIISA, these two states are first fed into an adaptive mask generator (AMG). AMG produces a pixelwise binary mask to quantify the differences between adjacent states. AMG begins by obtaining the differential representation $\triangle S^{norm}_{g\to ref} \in \mathbb{R}^{H\times W\times C}$ between S_g and S_{ref}, which can be expressed as

$$\triangle S^{norm}_{g\to ref} = \left|norm\left(S_g\right) - norm\left(S_{ref}\right)\right| \tag{15}$$

where $norm(\cdot)$ represents the normalization operation. Subsequently, a convolution and a softmax function are utilized to obtain the masking features $M_{g\to ref} \in \mathbb{R}^{H\times W}$ of S_g and S_{ref}, which can be described as

$$M_{g\to ref} = softmax\left(f_{conv}\left(\triangle S^{norm}_{g\to ref}\right)\right) \tag{16}$$

where $softmax(\cdot)$ is the softmax function, and f_{conv} represents a 1×1 2-D convolution. The masking features $M_{g\to ref}$ provide the probability of activation at each position. Finally, the final binary mask $Mask_{g,ref} \in \mathbb{R}^{H\times W}$ can be obtained by thresholding $M_{g\to ref}$ with a predefined value τ

$$Mask_{g,ref} = \begin{cases} 1, & \text{if } M_{g\to ref} > \tau \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

The above adaptive binary mask indicates whether a position should be activated or suppressed, which provides guidance for the following masked attention calculation.

Since our goal is to enhance the current spectral state by integrating the adjacent reference state, the intrastate query Q_{intra} is derived solely from the current state. Consequently, the interstate key K_{inter} and the value V_{inter} are generated from the concatenation of the current state S_g and the reference state S_{ref}

$$\begin{cases} Q_{intra} = f_{\text{proj}_q}\left([S_g]\right) \\ K_{inter} = f_{\text{proj}_k}\left([S_g, S_{ref}]\right) \\ V_{inter} = f_{\text{proj}_v}\left([S_g, S_{ref}]\right) \end{cases} \tag{18}$$

where the functions $f_{\text{proj}_q}(\cdot)$, $f_{\text{proj}_k}(\cdot)$, and $f_{\text{proj}_v}(\cdot)$ are the independent projection and reshaping operations for the query, key, and value, respectively, to form the final matrices $Q_{intra} \in \mathbb{R}^{HW\times D}$ and $K_{inter}, V_{inter} \in \mathbb{R}^{2HW\times D}$ used in the subsequent attention computation.

Unlike traditional attention mechanisms that compute relations across all spatial positions, MIISA adopts a more efficient select-then-attend strategy. It first utilizes the masking mechanism to identify a sparse set of core regions. For the query, only the positions with a mask value of 1 are involved in the attention calculation

$$A_{MIISA} = \text{softmax}\left(\frac{Q_s K^T_{inter}}{\sqrt{d_k}}\right) V_{inter} \tag{19}$$

where $Q_s \in \mathbb{R}^{\alpha HW\times D}$ represents the selected query, α is the ratio of selected regions of Q_{intra}, and d_k represents the dimensionality of the key vectors.

The proposed MIISA yields a dual benefit. Computationally, it reduces the number of query vectors involved in the attention calculation, thus lowering the overall computational cost. Qualitatively, by focusing attention exclusively on these core areas, it effectively suppresses interference from noisy or spectrally uncorrelated regions and ensures that only beneficial spatial textures are integrated for reconstruction.

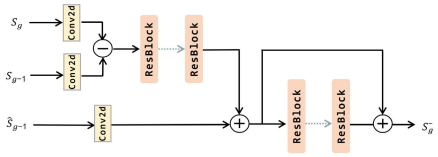


Fig. 2. Workflow of the proposed state predicting module.

3) State Predicting: In our framework, the refined state $\hat{S}_{g-1}$ produced by the ASPU for the previous spectral group often contains rich spatial details that may be blurred or missing in the current state S_g. However, as previous research [8] has revealed, adjacent spectral groups often exhibit significant differences in spectral response and noise distribution due to varying ground reflectance. This implies that while the previous state might offer beneficial spatial textures, it could simultaneously carry uncorrelated spectral information or noise. Therefore, a direct fusion of the current state with this prior information risks introduces detrimental interference.

To this end, we parameterize the state predicting module according to (4) to generate the prior estimation of the current state. State predicting perceives the discrepancies between the current state S_g and its predecessor S_{g-1} to correct the refined previous state $\hat{S}_{g-1}$, providing a more reliable reference for the subsequent state update. The details of state predicting are shown in Fig. 2. Specifically, it first conducts an explicit differential operation between the current state and the previous state. Next, several cascaded residual blocks are employed to enrich the perception of differences. The above procedures can be expressed as

$$\triangle S_{g\to g-1} = f_{res}\left(f_{conv_1}\left(S_g\right) - f_{conv_2}\left(S_{g-1}\right)\right) \tag{20}$$

where $\triangle S_{g\to g-1} \in \mathbb{R}^{H\times W\times C}$ represents the differential information between S_g and S_{g-1}. Subsequently, the previous refined state $\hat{S}_{g-1}$ is fused with the differential information through an addition operation, which enables correction of the differences in the adjacent state information. Finally, the posterior estimation of the current state can be obtained after several cascaded residual blocks. It contains the corrected information from the previous state. The aforementioned steps can be described as follows:

$$S^-_g = f_{res}\left(f_{conv}\left(\hat{S}_{g-1}\right) + \triangle S_{g\to g-1}\right) \tag{21}$$

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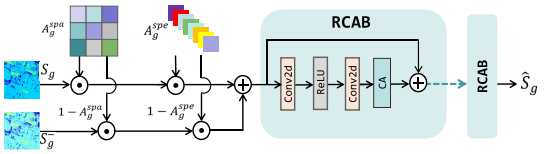


Fig. 3. Workflow of the proposed SU module.

where $S_g^- \in \mathbb{R}^{H\times W\times C}$ represents the prior estimation of the current state.

4) State Updating: After obtaining the prior estimation S_g^- of the current state, the SU module is parameterized according to (5) to adaptively activate information from the prior and update the current state. The workflow of SU is illustrated in Fig. 3. The update is performed based on the adaptive activation mask. High mask values indicate high-confidence regions, leading to the preservation of original features from the current state. Conversely, low mask values identify regions where spatial details may be missing. In these areas, the mechanism activates the reliable textures from the high-quality prior S_g^- to restore and enhance the current state. It can be expressed as follows:

$$S_g^+ = A_g^{spa} \cdot A_g^{spe} \cdot S_g + \left(1 - A_g^{spa}\right) \cdot \left(1 - A_g^{spe}\right) \cdot S_g^- \tag{22}$$

where S_g^+ represents the updated state. In order to further explore spatial and spectral information, several cascaded residual CA blocks (RCABs) proposed in [28] are employed to further enhance spatial details and spectral consistency. The final refined state can be obtained by

$$\hat{S}_g = \mathcal{RCAB}_n\left(\ldots \mathcal{RCAB}_1\left(S_g^+\right)\ldots\right) \tag{23}$$

where $\hat{S}_g \in \mathbb{R}^{H\times W\times C}$ represents the refined state obtained from the ASPU.

Due to the proposed ASPU, our method could adaptively activate information from the previous state based on difference perception to refine the current state. It effectively utilizes complementary spectral information without introducing irrelevant information.

D. Bidirectional Cross-State Alignment

Although existing recurrent-based methods attempt to utilize interband context to assist the reconstruction of the current band, they still face an inherent dilemma regarding information flow. As mentioned above, unidirectional methods can only propagate information from previous bands to the current band for reconstruction, which may lead to an imbalance of information between different spectral bands. Although bidirectional methods can fully utilize the bidirectional context, they often incur significant computational costs.

To overcome this dilemma, we propose a novel BCSA strategy to achieve the benefits of bidirectional context while mitigating the associated costs. The architecture of BCSA is shown in Fig. 1(b). The BCSA utilizes cached representations of back-and-forth states to further align the refined current state $\hat{S}_g$ obtained from the ASPU. Specifically, the BCSA operates based on a dynamic caching mechanism. As the network processes the sequence along the spectral dimension, each state $\hat{S}_g$ refined by the ASPU is temporarily stored in a cache. This cached state is then utilized in two ways: it is passed forward to update the back cache of the subsequent state $\hat{S}_{g+1}$ within the BCSA, and it is propagated backward to update the forth cache of the preceding state $\hat{S}_{g-1}$. Once $\hat{S}_g$ has fulfilled its role in aligning the adjacent states (i.e., after the processing of $\hat{S}_{g+1}$ is completed), it is released from the cache as the caches are updated by the newly refined states. This "just-in-time" caching strategy ensures that bidirectional context is always available with minimal memory overhead.

For the alignment of $\hat{S}_g$, the proposed BCSA first obtains adjacent states $\hat{S}_b \in \mathbb{R}^{H\times W\times C}$ and $\hat{S}_f \in \mathbb{R}^{H\times W\times C}$ from the cached past and future states, respectively. In order to fully utilize bidirectional information, we employ a dual-branch structure to separately process the interaction of the current refined state $\hat{S}_g$, obtained from the ASPU, with both the back-and-forth cached information. The two branches are responsible for modeling the information flows from back ($\hat{S}_b \to \hat{S}_g$) and forth ($\hat{S}_f \to \hat{S}_g$), respectively. This dual-branch design, powered by the "just-in-time" cache, allows the model to access both past and future contexts simultaneously within a purely unidirectional dataflow. This is the key to achieving bidirectional alignment without the computational overhead of a traditional bidirectional structure.

As discussed previously, it is desirable to perceive both the differences and the correlations between adjacent states when aggregating their information. To achieve this within BCSA, the MIISA is employed to achieve efficient information interaction between adjacent states and suppress the introduction of irrelevant information through the adaptive mask mechanism. This attention-based fused representation is then further processed by a residual block $f_{res}(\cdot)$ to obtain the final fusion result. Taking the branch of $\hat{S}_b$ as an example, the fusion result $S_{g,b}^{fuse} \in \mathbb{R}^{H\times W\times C}$ can be obtained by

$$S_{g,b}^{fuse} = f_{res}\left(f_{MIISA}\left(f_{conv}\left(\hat{S}_g\right), f_{conv}\left(\hat{S}_b\right)\right)\right) \tag{24}$$

where $f_{MIISA}(\cdot)$ is the proposed MIISA. Afterward, we adopt a differential operation to extract the effective complementary information $\triangle S_{g\to b} \in \mathbb{R}^{H\times W\times C}$ for this branch

$$\triangle S_{g\to b} = f_{res}\left(f_{res}\left(f_{conv}\left(\hat{S}_g\right)\right) - S_{g,b}^{fuse}\right). \tag{25}$$

The complementary information $\triangle S_{g\to f}$ from the forward branch is obtained through a similar process. The bidirectional complementary information from two branches could effectively enhance and align the current state. Finally, these bidirectional complementary components are integrated with the refined current state to acquire the aligned state

$$S_g^{align} = f_{conv}\left(f_{res}\left(f_{conv}\left(\hat{S}_g\right)\right) + \triangle S_{g\to b} + \triangle S_{g\to f}\right) \tag{26}$$

where $S_g^{align} \in \mathbb{R}^{H\times W\times C}$ is the aligned state of the spectral groups, which serves as the final output of the BCSA.

Through its "just-in-time" caching strategy and the absence of a full backward pass, the proposed BCSA achieves high computational efficiency while enabling the propagation of bidirectional information.

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E. Loss Function

In existing studies, most SR methods employ pixelwise loss (e.g., L_1 loss) to achieve a high PSNR. In this work, we adopt the L_1 loss as our content loss to measure the accuracy of the network, which can be expressed as

$$L_{con} = \|I^{sr} - I^{hr}\|_1 \tag{27}$$

where I^{sr} and I^{hr} are the reconstructed SR image and the corresponding ground-truth HR image, respectively.

The above content loss focuses on preserving spatial information but overlooks the spectral consistency in HSIs, which may lead to spectral distortion. To better maintain spatial–spectral consistency, we incorporate a spectral loss similar to [20] to further constrain the network. The spectral loss can be formulated as follows:

$$L_{spe} = \frac{1}{\pi} arccos\left(\frac{I^{hr} \cdot I^{sr}}{\|I^{hr}\|_2 \|I^{sr}\|_2}\right). \tag{28}$$

Previous studies [7], [20] suggest that gradient maps can better reflect the spatial–spectral dependence of HSIs. Thus, we introduce a gradient loss to improve the sharpness of reconstructed images in both spatial and spectral domains, which is denoted as

$$L_{gra} = \left\|\mathcal{G}\left(I^{sr}\right) - \mathcal{G}\left(I^{hr}\right)\right\|_1 \tag{29}$$

where $\mathcal{G}(\cdot)$ is the operator to obtain the gradient value along the horizontal, vertical, and spectral dimensions of the image.

Finally, the total loss for the proposed network can be defined as

$$L = L_{con} + \lambda_1 L_{spe} + \lambda_2 L_{gra} \tag{30}$$

where λ_1 and λ_2 are the tradeoff parameters that control the weights of the different loss terms. In our experiments, we set the tradeoff parameters as $\lambda_1 = 0.5$ and $\lambda_2 = 0.1$.

IV. Experiments and Results

A. Datasets and Experimental Setup

In our experiments, we adopt three benchmark HSI datasets, i.e., Houston dataset, Chikusei dataset [59], and Pavia dataset [60] to evaluate our method. We compare the proposed method against eight state-of-the-art (SOTA) methods, including ERCSR [41], DualSR [49], GELIN [6], AS^{3}ITransUnet [61], MSDFormer [7], SRDNet [42], HyperSIGMA [52], and MambaHSISR [48]. In addition, the bicubic interpolation is selected as the baseline.

To comprehensively evaluate the performance of our method in terms of both spatial fidelity and spectral consistency, we employ four evaluation metrics: the PSNR, structural similarity (SSIM), spectral angle mapper (SAM), and cross correlation (CC). The optimal values for these metrics are $+\infty$, 1, 0, and 1, respectively. For PSNR, SSIM, and CC, higher values indicate better performance, while for SAM, a lower value is preferred.

TABLE I

Quantitative Comparison of SOTA Methods at Different Scales on Houston Dataset. The Best Results are Marked in Bold, and the Second-Best Results are Marked in Underline

Methods	Scale	PSNR↑	SSIM↑	SAM↓	CC↑
Bicubic		42.172	0.9655	2.468	0.9790
ERCSR [41]		44.042	0.9756	2.200	0.9859
DualSR [49]		44.238	0.9767	2.108	0.9868
GELIN [6]	×4	44.397	0.9776	1.960	0.9871
AS^{3}ITransUNet [61]		44.279	0.9770	2.041	0.9867
MSDFormer [7]		44.327	0.9771	1.926	0.9869
SRDNet [42]		44.465	0.9778	1.962	0.9871
HyperSIGMA [52]		44.409	0.9776	1.950	0.9873
MambaHSISR [48]		44.504	0.9779	1.953	0.9873
DRSN(Ours)		**44.695**	**0.9789**	**1.841**	**0.9879**
Bicubic		37.154	0.9192	4.492	0.9335
ERCSR [41]		38.354	0.9302	4.306	0.9472
DualSR [49]		38.515	0.9320	4.152	0.9488
GELIN [6]	×8	38.768	0.9360	3.721	0.9519
AS^{3}ITransUNet [61]		38.579	0.9334	3.672	0.9505
MSDFormer [7]		38.598	0.9339	3.658	0.9506
SRDNet [42]		38.747	0.9351	3.700	0.9518
HyperSIGMA [52]		38.799	0.9360	**3.542**	0.9522
MambaHSISR [48]		38.642	0.9324	3.694	0.9500
DRSN(Ours)		**38.851**	**0.9364**	3.578	**0.9526**

B. Implementation Details

In the proposed DRSN, the input HSI is first divided into G nonoverlapping groups along the spectral dimension. Each group consists of n spectral bands, with the final group zero-padded to match this size if needed. In our experiments, we set $n = 4$ for Houston dataset, and $n = 8$ for Pavia and Chikusei datasets. In the training phase, we empirically set the batch size to 32 and use the Adam optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) to train the network for 300 epochs. The initial learning rate is set to 1×10^{-4} and halved every 60 epochs. The proposed network is implemented using the PyTorch framework and trained on an NVIDIA GeForce RTX 3090 GPU.

C. Comparisons With the SOTA Methods

1) Experimental Results on Houston Dataset: Houston 2018 dataset is an image with spatial dimensions of 4172 × 1202 pixels, captured by the ITRES CASI 1500 spectral imager over the University of Houston campus. It consists of 48 bands in the spectral range of 380–1050 nm. Following the setup in [16], the bottom of the image is cropped into four nonoverlapping patches, each with a spatial resolution of 512 × 512 pixels, which are used as the test set. The remaining part of the image is utilized to construct the training set by extracting patches. Specifically, for scale factors of 4 and 8, we randomly crop patches of 64 × 64 and 128 × 128 sizes as the corresponding HR ground-truth images, respectively. The corresponding LR HSIs are generated by bicubic downsampling.

To quantitatively evaluate the performance of the proposed DRSN, we compare it with six SOTA methods on Houston dataset. Table I summarizes the average values of four evaluation metrics at scale factors of 4 and 8. Although ERCSR [41], SRDNet [42], and MambaHSISR [48] effectively restore spatial information, they lack competitiveness in terms of

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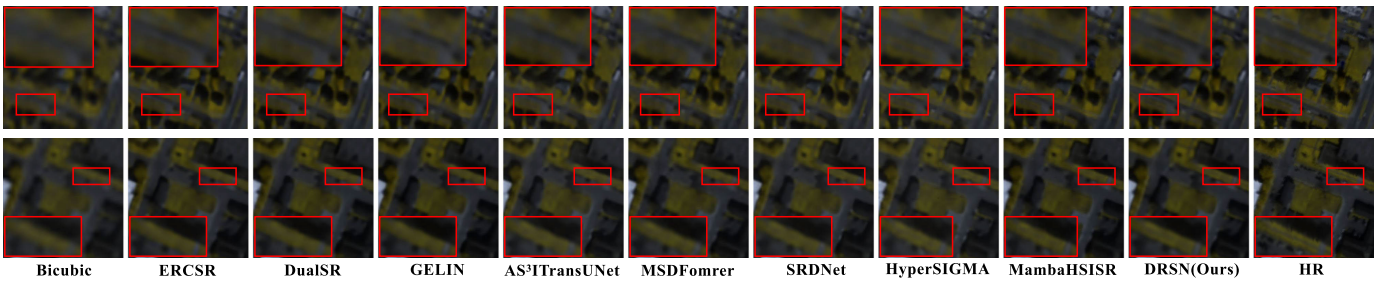


Fig. 4. Visualization results of two parts of the test image in Houston dataset with spectral bands 29–26–19 as R–G–B at a scale factor of 4.

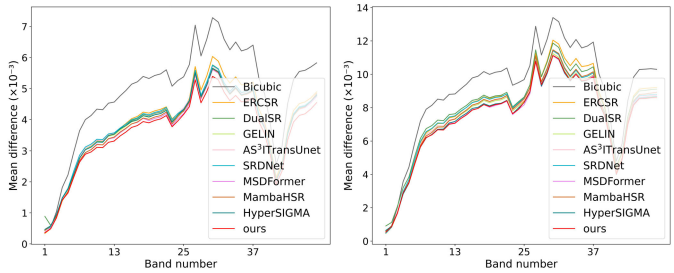


Fig. 5. Mean absolute error curves of test images on Houston dataset at scale factors of (left) 4 and (right) 8.

spectral similarity. DualSR [49], GELIN [6], AS^{3}TransUNet [61], and MSDFormer [7] exploit the similarity of spectral bands through grouping strategies, but their exploration of spectral differences is limited, potentially leading to suboptimal results. As shown in Table I, DRSN achieves superior performance across all metrics at both scale factors. Particularly at a scale factor of 4, our method exhibits a clear superiority over the other competing methods, which demonstrates that DRSN offers both accurate spatial restoration and superior spectral fidelity.

In order to further demonstrate the advantages of the proposed DRSN, we visualize the reconstruction results of two representative test images at a scale factor of 4 in Fig. 4. Specifically, the 29th, 26th, and 19th bands of the HSI are selected as RGB channels for better visualization. As shown in the figure, the visualization results indicate that DRSN achieves superior visual reconstruction quality with more accurate recovery of details. In terms of spectral fidelity, Fig. 5 presents the mean absolute spectral error curves of all test images at scale factors of 4 and 8 using different methods. It can be observed that DRSN achieves performance comparable to that of the second-best method in the initial bands, while its advantage becomes increasingly pronounced in the subsequent bands. This is mainly attributed to the proposed ASPU, which utilizes previous bands as reference information to effectively aggregate complementary information across spectral groups. The curve corresponding to the proposed method is the lowest at both scale factors, indicating superior spectral fidelity. Compared with other methods, DRSN models the HSI as a dynamic system evolving along the spectral dimension. Due to the proposed ASPU mechanism, our method effectively explores interband correlations and fully leverages spectral complementary information. Furthermore, our method avoids spectral distortion through the proposed BCSA, which allows it to achieve results with high spatial–spectral fidelity.

TABLE II

QUANTITATIVE COMPARISON OF SOTA METHODS AT DIFFERENT SCALES ON THE CHIKUSEI DATASET. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE MARKED IN UNDERLINE

Methods	Scale	PSNR↑	SSIM↑	SAM↓	CC↑
Bicubic		36.493	0.8971	3.362	0.9696
ERCSR [41]		38.038	0.9259	2.897	0.9772
DualSR [49]		37.423	0.9185	3.462	0.9777
GELIN [6]	×4	38.241	0.9315	2.597	0.9816
AS^{3}ITransUNet [61]		38.183	0.9306	2.600	0.9805
MSDFormer [7]		38.306	0.9326	2.565	0.9821
SRDNet [42]		38.234	0.9307	2.713	0.9817
HyperSIGMA [52]		38.490	0.9321	2.681	0.9823
MambaHSISR [48]		38.431	0.9301	2.726	0.9787
DRSN(Ours)		**38.579**	**0.9331**	**2.538**	**0.9833**
Bicubic		33.294	0.8231	5.008	0.9358
ERCSR [41]		33.920	0.8425	4.694	0.9425
DualSR [49]		33.432	0.8306	5.521	0.9448
GELIN [6]	×8	34.133	0.8537	4.211	0.9529
AS^{3}ITransUNet [61]		34.077	0.8465	4.455	0.9527
MSDFormer [7]		34.224	0.8545	4.167	**0.9542**
SRDNet [42]		34.173	0.8511	4.364	0.9537
HyperSIGMA [52]		34.404	0.8565	4.149	0.9535
MambaHSISR [48]		34.034	0.8508	4.522	0.9476
DRSN(Ours)		**34.446**	**0.8571**	**4.094**	0.9537

2) Experimental Results on the Chikusei Dataset: The Chikusei dataset [59], acquired by the Headwall Hyperspec-VNIR-C sensor, contains 2517 × 2335 pixels and 128 bands with a spectral range of 363–1018 nm. Due to the presence of invalid information at the edges of the original image, we crop a central region of 2048 × 2048 pixels from the original image and further divide it into training and test sets. The same strategies for dataset division and patch cropping as those employed for Houston dataset are adopted for the Chikusei dataset.

Table II presents the average values of the evaluation metrics for all methods on the Chikusei dataset at scale factors of 4 and 8. In contrast to Houston dataset, the Chikusei dataset contains more spectral bands and a higher level of noise, which increases the difficulty of reconstruction. As shown in Table II, the proposed DRSN outperforms all other methods on most of the evaluated metrics. Particularly at a scale factor of 4, DRSN exhibits a clear superiority over the other competing methods. This indicates that DRSN is better equipped to handle challenging hyperspectral datasets.

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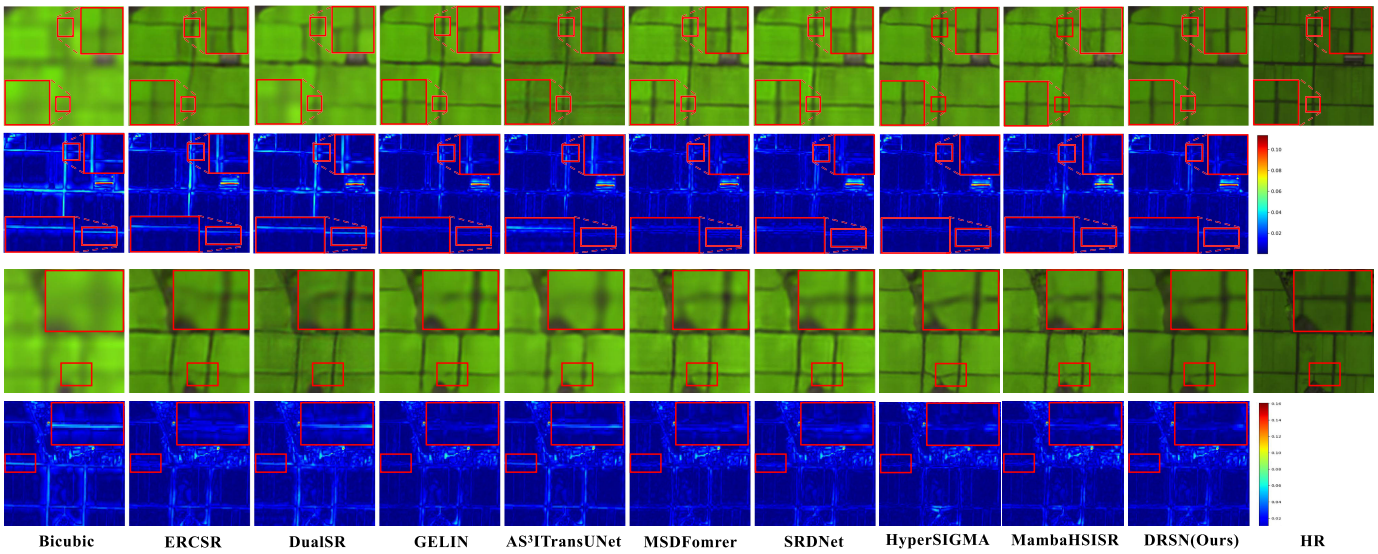


Fig. 6. Visualization results of two parts of the test image in the Chikusei dataset with spectral bands 70–100–36 as R–G–B at a scale factor of 8. The corresponding mean absolute error maps are shown in the second row.

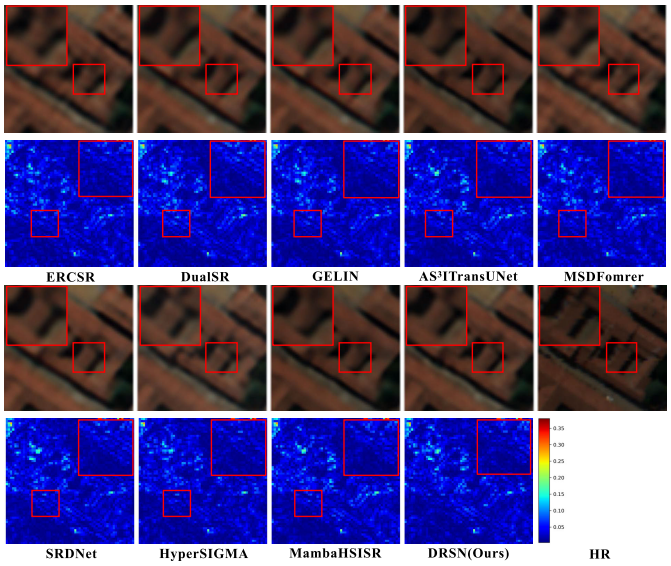


Fig. 7. Visualization results of a part of the test image in Pavia dataset with spectral bands 60–31–12 as R–G–B at a scale factor of 4. The corresponding mean absolute error maps are shown in the second row.

For a more diverse visualization and without loss of generality, we present the challenging scale $\times 8$ results on the Chikusei dataset in Fig. 6. Specifically, we select the 70th, 100th, and 36th bands of the HSI as RGB channels. The proposed DRSN successfully restores more detailed information while minimizing the appearance of artifacts. The superiority of our DRSN is further highlighted by examining the magnified details within the red box. In addition, the mean absolute error maps for the visualized images are also shown in Fig. 6. As can be seen, DRSN achieves a lower error level, which demonstrates that the proposed method still performs better in restoring spatial details even when facing larger scale factors. This is primarily attributed to the ability of the proposed ASPU

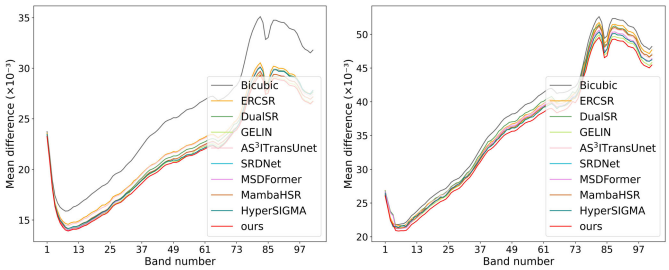


Fig. 8. Mean absolute error curves of test images on Pavia dataset at scale factors of (left) 4 and (right) 8.

to effectively leverage spectral complementary information to recover missing details.

3) Experimental Results on Pavia Dataset: Pavia Center dataset [60] was captured by the ROSIS sensor during a flight campaign over the center area of Pavia, northern Italy, in 2001. It contains 1096 $\times$ 1096 pixels with 102 bands in the spectral range of 430–860 nm. Due to the invalid region, we crop the informative region with $1096 \times 715 \times 102$ of the original scene. For Pavia dataset, we adopt the same strategies for dataset division and cropping as those employed for Houston dataset.

Table III reports the average values of four evaluation metrics for all methods on Pavia dataset. Compared with the Chikusei and Houston datasets, Pavia dataset provides fewer training samples. As shown in Table III, our method consistently outperforms other competing methods in all metrics at both scale factors. This indicates that our method remains more competitive than other methods even when dealing with limited training data. To provide qualitative analysis, we visualize the reconstruction results of test images from Pavia dataset in Fig. 7. Specifically, we select the 30th, 20th, and 12th bands of the HSI as RGB channels. The figure also includes the corresponding mean absolute error maps for each reconstructed image. It can be observed that our method recovers clearer details and achieves minimum error levels compared to other methods. In addition, Fig. 8 shows

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TABLE III

QUANTITATIVE COMPARISON OF SOTA METHODS AT DIFFERENT SCALES ON PAVIA DATASET. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE MARKED IN UNDERLINE

Methods	Scale	PSNR↑	SSIM↑	SAM↓	CC↑
Bicubic	×4	28.081	0.7403	6.455	0.9123
ERCSR [41]		29.103	0.7915	5.969	0.9302
DualSR [49]		29.152	0.7964	5.935	0.9311
GELIN [6]		29.239	0.7997	5.788	0.9326
AS^{3}TransUNet [61]		29.189	0.7997	5.913	0.9318
MSDFormer [7]		29.159	0.7943	5.888	0.9312
SRDNet [42]		29.242	0.8030	5.798	0.9329
HyperSIGMA [52]		29.259	**0.8091**	5.816	0.9332
MambaHSISR [48]		29.189	0.7997	5.781	0.9318
DRSN(Ours)		**29.320**	0.8050	**5.766**	**0.9340**
Bicubic	×8	24.995	0.5611	8.209	0.8146
ERCSR [41]		25.180	0.5787	8.276	0.8216
DualSR [49]		25.164	0.5700	8.274	0.8204
GELIN [6]		25.303	0.5805	8.106	0.8259
AS^{3}TransUNet [61]		25.225	0.5822	8.240	0.8235
MSDFormer [7]		25.240	0.5808	8.193	0.8238
SRDNet [42]		25.299	0.5872	7.988	0.8273
HyperSIGMA [52]		25.309	0.5898	8.026	0.8304
MambaHSISR [48]		25.225	0.5822	8.240	0.8235
DRSN(Ours)		**25.428**	**0.5920**	**7.973**	**0.8320**

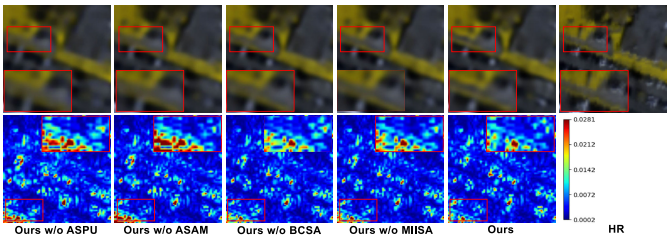


Fig. 9. Ablation study visualization on Houston dataset (×4).

the average absolute spectral error curves for all test images on Pavia dataset at scale factors of 4 and 8. It is clear that the mean spectral error of DRSN is the lowest, which indicates that DRSN can also achieve the best performance when handling HSIs with a larger number of spectral bands.

D. Ablation Study

In this section, we analyze the impact of different components on performance through several ablation studies. Unless otherwise specified, all experiments are conducted on Houston dataset at a scale factor of 4.

1) Effectiveness of ASPU: In our DRSN, the proposed ASPU module is employed to adaptively activate the cross-state complementary information to align and enhance the current state. To verify the effectiveness of ASPU, we remove all components from the ASPU except for the cascaded RCABs. Consequently, the state information is propagated and processed solely by the remaining cascaded RCABs without the ASAM, state predicting, and state updating provided by ASPU. This variant is denoted as "Ours w/o ASPU" in Table IV. A visual comparison is shown in Fig. 9. It is evident that the absence of ASPU leads to a substantial decrease in both PSNR and SSIM, resulting in a loss of details.

In addition, the ASAM is the core component of the ASPU, which adaptively activates complementary information through spatial and spectral activation masks. To demonstrate the effectiveness of ASAM, we remove the ASAM and directly fuse the current and previous states with an addition operation. As shown in Table IV, the variant "Ours w/o ASAM" shows a decrease in all evaluation indices compared to the original model "Ours." A visual comparison is also provided in Fig. 9. These results demonstrate that the proposed ASAM enables the selective utilization of complementary information while suppressing the inclusion of irrelevant content.

TABLE IV

ABLATION STUDIES ON COMPONENTS. QUANTITATIVE COMPARISONS AMONG SOME VARIANTS ON HOUSTON DATASET AT A SCALE FACTOR OF 4

Model	PSNR↑	SSIM↑	SAM↓
Ours w/o ASPU	44.246	0.9767	1.984
Ours w/o ASAM	44.497	0.9779	1.905
Ours w/o BCSA	44.539	0.9782	1.883
Ours w/o MIISA	44.611	0.9786	1.856
Ours	**44.695**	**0.9789**	**1.841**

2) Effectiveness of BCSA: BCSA utilizes the cache of back-and-forth states to further align and enhance the current state, which partially alleviates the issue of information imbalance caused by the unidirectional structure. To evaluate the effectiveness of BCSA, we remove the BCSA at the end of the network. The resulting variant is denoted as "Ours w/o BCSA" in Table IV. As shown in the table, after removing the BCSA, the performance of our network decreases in all aspects. For example, the PSNR of the proposed method is nearly 0.16 dB higher than the variant "ours w/o BCSA," and the SAM is lower by 0.042. As shown in Fig. 9, the removal of BCSA leads to slightly blurred details. The above experimental results clearly demonstrate the effectiveness of the proposed BCSA. In general, BCSA enables our network to achieve the efficiency of unidirectional networks while enabling bidirectional information propagation.

3) Effectiveness of MIISA: The MIISA enables efficient inter- and intraspectral information fusion and suppresses the introduction of irrelevant information via the adaptive mask. To evaluate the effectiveness of MIISA, we replace it with a 2-D convolution to fuse the concatenated adjacent spectral features. This variant is denoted as "Ours w/o MIISA" in Table IV. We also show the visual results in Fig. 9. As shown in the table, replacing MIISA with a simpler 2-D convolution results in a decline across all evaluation metrics. For instance, the PSNR of "Ours w/o MIISA" decreases by nearly 0.08 dB compared to "Ours." This is because there are complex correlations between adjacent spectral bands, and simple concatenation and 2-D convolution cannot effectively facilitate information interaction between them. The introduction of MIISA enables the selection of complementary information through the adaptive mask, thereby efficiently fusing information from adjacent bands.

4) Effect of Loss Functions: We report the results under different combinations of losses in Table V. When only L_{con} is used, DRSN achieves a relatively high PSNR but a poor SAM. This is because L_{con} focuses on spatial information while

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TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT LOSS FUNCTIONS ON HOUSTON DATASET AT A SCALE FACTOR OF 4

L_{con}	L_{spe}	L_{gra}	PSNR↑	SSIM↑	SAM↓
✓			44.684	0.9787	1.958
✓	✓		44.626	0.9786	1.881
✓		✓	44.689	0.9788	1.925
✓	✓	✓	**44.695**	**0.9789**	**1.841**

TABLE VI

IMPACT OF DIFFERENT GROUP SIZES (n) ON PSNR (DB) AND FLOPS (G)

Dataset	$n=2$		$n=4$		$n=8$		$n=12$	
	PSNR	FLOPs	PSNR	FLOPs	PSNR	FLOPs	PSNR	FLOPs
Houston	44.654	387	44.695	190	44.374	97	44.279	61
Chikusei	38.501	1197	38.553	600	38.579	302	38.486	209
Pavia	29.314	953	29.311	487	29.320	245	29.309	170

TABLE VII

GENERALITY ASSESSMENT OF THE PROPOSED DRSN. THE MODEL IS EVALUATED ON HOUSTON TEST SET AFTER BEING TRAINED ON HOUSTON DATASET AND THE CHIKUSEI DATASET

Methods	Scale	PSNR↑	SSIM↑	SAM↓	CC↑
Ours (H)	×4	44.695	0.9789	1.841	0.9879
Ours (C)		44.087	0.9759	2.011	0.9861
Ours (H)	×8	38.851	0.9364	3.578	0.9526
Ours (C)		38.134	0.9286	4.132	0.9456

neglecting the spectral consistency. While slightly decreasing the PSNR, the inclusion of the spectral loss L_{spe} leads to a significant improvement in SAM. This indicates that the introduction of the L_{spe} preserves spectral consistency and reduces spectral distortion. Afterward, the gradient loss L_{gra} is further introduced to sharpen the structural information, which can balance L_{con} and L_{spe}. The best performance is achieved when all three items are utilized.

5) Effect of Group Size n: The choice of n balances the exploitation of intragroup complementarity with the adequacy of intergroup evolution. Specifically, an excessively large n complicates intragroup features and reduces evolution steps, limiting self-refinement. Conversely, a small n yields a sparse context and high computational cost. As shown in Table VI, for Houston dataset, the optimal tradeoff is achieved when n is set to 4. For Chikusei and Pavia, setting n to 8 maintains better performance with lower floating-point operations (FLOPs) compared to the case of n = 4.

E. Analysis of Parameter and Complexity

In this section, we analyze the model size and computational cost. Fig. 10 illustrates the number of FLOPs, the number of parameters, and the PSNR of different methods. The FLOPs of all methods are calculated using an input size of $64\times64\times48$ at the scale factor of 4. Due to the significant computational cost introduced by 3-D convolutions, the FLOPs of ERCSR [41] and DualSR [49] are significantly higher. Although SRDNet [42] has lower FLOPs, it does not fully leverage spectral similarity, which may lead to suboptimal results. GELIN

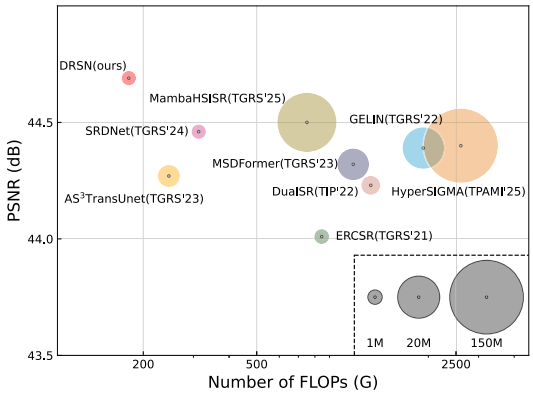


Fig. 10. Comparison of tradeoffs between model performance and efficiency on Houston dataset at a scale factor of 4. FLOPs are plotted on the horizontal axis, PSNR is represented on the vertical axis, and the area of each circle is proportional to the number of parameters.

[6] and MSDFormer [7] leverage complementary information from adjacent spectral bands through grouping strategies, yet still suffer from large model sizes and FLOPs. Although HyperSIGMA [52] and MambaHSISR [48] achieve competitive results, they are constrained by their large model sizes. In our DRSN, we reformulate the HSIs from a nonlinear dynamic system perspective to fully leverage spectral similarity and complementarity. In addition, DRSN achieves the efficiency of unidirectional networks while enabling bidirectional information propagation through the proposed BCSA. As a result, our method achieves the best performance with the lowest FLOPs and a compact model size.

F. Analysis of Cross-Dataset HSISR

Hyperspectral datasets typically vary in the number of spectral bands. Most existing methods can only handle data with a fixed number of spectral bands, as those in the training set, thus failing to process data with an arbitrary spectral dimension. This limitation significantly curtails their practical applicability across diverse hyperspectral sensors and datasets. In this work, the proposed DRSN models HSIs as nonlinear dynamic systems and treats each spectral group as a state variable of the system. Since dynamic systems do not impose restrictions on the length of state variable sequences, our method can handle HSI data with an arbitrary spectral dimension. Compared to other methods, our approach offers superior generality and flexibility.

To verify the cross-dataset generalization of the proposed DRSN, we design two sets of comparative experiments at scale factors of 4 and 8. First, DRSN is trained on Houston dataset, establishing a baseline, and is evaluated on the corresponding test set, which is denoted as "Ours (H)" in Table VII. Subsequently, DRSN is trained on the Chikusei dataset and tested on Houston dataset. It is denoted as "Ours (C)" in Table VII. It is worth noting that the number of spectral bands in the HSIs of Houston and Chikusei datasets is different, specifically 48 and 128, respectively. Due to the difference in spectral bands, all the compared methods could not achieve the above cross-dataset HSISR task. As can be seen in Table VII, "Ours

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(C)" achieves results comparable to "Ours (H)" at both scale factors. This indicates that when the spectral bands of the test data differ from those of the training data, the proposed method can still achieve acceptable results. Compared with other methods, our method is capable of handling HSIs with an arbitrary number of spectral bands, which is a significant advantage over many existing HSISR methods that are often constrained to a fixed number of input bands determined during training. The above experiments demonstrate that our method exhibits strong generality.

V. Conclusion

In this article, we propose a novel HSISR framework, DRSN, which models HSIs as a nonlinear dynamic system evolving along the spectral dimension. This formulation enables our method to flexibly process HSIs with an arbitrary number of spectral bands, thereby overcoming the application limitations of existing methods. In addition, we introduce the ASPU module, which leverages spectral correlation to dynamically refine the spectral state through adaptive spatial–spectral activation guided by uncertainty estimation. Furthermore, we develop a BCSA module, which employs a back-and-forth caching strategy and MIISA to enable efficient bidirectional information propagation without excessive memory requirements. Extensive experiments on multiple hyperspectral datasets demonstrate that our method achieves SOTA performance with low parameters and computational costs.

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